Deep Learning — Group Project Guidelines

Practical work (100 % of the grade)

Annika Stiehl

1 In a nutshell

In groups of **3–4 students** you pick a **NeurIPS 2025** paper and work on **CHB-MIT**, **SeizeIT2** or an **image dataset**. You compare three models: **(A)** a simple baseline, **(B)** the paper's idea added to A or the paper's full method, and **(C)** B plus **your own idea**. You evaluate all three rigorously, publish your **code**, write a **6-page short paper** and present a **poster** on day 8 (**11 November 2026**). The project trains exactly what researchers do every day: *build a fair baseline, test whether a published idea transfers to your data, improve it, and report honestly.*

2 Steps

1. **Paper selection.** Choose a paper accepted at NeurIPS 2025 (main track or Datasets & Benchmarks track) that uses — or whose method can be applied to — one of the course datasets (Section 3).
2. **Dataset understanding and preprocessing.** Obtain the data, analyse it (class balance, recording lengths, channels, artifacts / image statistics) and document every preprocessing step.
3. **A — Baseline.** A simple standard model for your task and data, e.g. from the course labs (band power + MLP, 1D CNN, ResNet-18). Same preprocessing, split and evaluation as B and C.
4. **B — Paper.** Add the paper's key idea to A, *or* run the paper's full method (e.g. its code) on your data. Both are allowed; say which in the proposal. *Optional:* reproduce the paper's published numbers on its own data as far as your compute allows, and explain the gap.
5. **C — Your idea.** Propose and implement a *motivated* change to B (Section 4) and an ablation that shows its effect.
6. **Evaluation.** Compare A, B and C on a held-out test set with suitable metrics and statistical tests: B vs. A (does the paper's idea help on your data?) and C vs. B (does your idea help?) (Section 5).
7. **Code, paper, poster** (Section 6).

How to find NeurIPS 2025 papers

NeurIPS 2025 is complete: all accepted papers, their reviews and the official proceedings are public. Main track and Datasets & Benchmarks track both count.

- **Proceedings** (PDFs, BibTeX): https://proceedings.neurips.cc/paper_files/paper/2025
- **List of accepted papers** with search: https://neurips.cc/virtual/2025/papers.html; as JSON (title, abstract, OpenReview link): https://neurips.cc/static/virtual/data/neurips-2025-orals-posters.json. Search it in Colab:

```
import pandas as pd, requests; URL = "<JSON link above>"
df = pd.DataFrame(requests.get(URL).json()["results"])
df[df.name.str.contains("EEG|seizure", case=False)][["name", "paper_url"]]
```

- **OpenReview** (paper, reviews, discussion with the authors): https://openreview.net/group?id=NeurIPS.cc/2025/Conference and https://openreview.net/group?id=NeurIPS.cc/2025/Datasets_and_Benchmarks_Track. The reviews are worth reading: they often name the weaknesses that make a good idea for C.

- **Code**: link on the paper page or GitHub search for the paper title / "NeurIPS 2025".
- A good paper for this project has: a public dataset that matches (or a method that transfers), **public code**, an idea you can add to a simple baseline (or a method you can run on your data), a training run of at most $\approx$ 20 min on a Colab T4 GPU (you need A, B and C with several seeds each) (or a released checkpoint you can fine-tune; never pretrain a foundation model yourself), and a method you can explain after one careful reading.
- The paper must be approved by the lecturer (proposal, day 3). Two groups may not pick the same paper.

Starting points (checked on 13 Oct 2026 — verify code availability yourself)

Paper (NeurIPS 2025)	Topic	Notes
EvoBrain: dynamic multi-channel EEG graph modeling (https://openreview.net/forum?id=XmV7KRABBl)	seizure detection, TUSZ + **CHB-MIT**	code: https://github.com/Kotoge/EvoBrain; runs directly on CHB-MIT
FAPEX: cross-subject seizure prediction (https://openreview.net/forum?id=A9jXG3FUMT)	seizure *prediction*	code not found yet — check the paper page; the amplitude-phase idea can be added to a simple CNN (B)
LUNA: topology-agnostic EEG foundation model (https://openreview.net/forum?id=uazfjnFL0G)	EEG with different channel layouts	code: https://github.com/pulp-bio/biofoundation; efficient; scalp $\rightarrow$ wearable (SeizeIT2)
REVE: EEG foundation model, 25 000 subjects (https://openreview.net/forum?id=ZeFMtRBy4Z)	many EEG tasks	fine-tune or linear-probe released weights, never pretrain; check weights / code on the paper page
Is limited participant diversity impeding EEG-based ML? (https://openreview.net/forum?id=cWn3RXJQ7G)	data / evaluation study	code: https://github.com/bomatter/participant-diversity-paper; question for CHB-MIT: how many patients do you need?
BrainOmni: foundation model for EEG and MEG (https://openreview.net/forum?id=cjHQj0tCy6)	EEG / MEG	code: https://github.com/OpenTSLab/BrainOmni

Image projects: search the list for keywords such as MedMNIST, CIFAR, few-shot, parameter-efficient fine-tuning or robustness.

3 Datasets

	Content & access	Notes
CHB-MIT	Scalp EEG of 22 paediatric patients (23 cases, Boston Children's Hospital), mostly 23 bipolar channels, 256 Hz, 198 seizures. Open access on PhysioNet: https://physionet.org/content/chbmit/1.0.0/; fast mirror: https://physionet-open.s3.amazonaws.com/chbmit/1.0.0/	Seizure times in `chbXX-summary.txt`. Montage changes in some patients — check channel names. Patient-specific and cross-patient settings are both common in the literature.
SeizeIT2	Wearable behind-the-ear EEG (2 channels, 256 Hz) + ECG, EMG, accelerometer/gyroscope of 125 patients with focal epilepsy from 5 European centres, $\approx$ 11 640 h, 886 seizures. BIDS format, CC0, OpenNeuro `ds005873`: https://openneuro.org/datasets/ds005873; direct download: https://s3.amazonaws.com/openneuro.org/ds005873/. Paper: Bhagubai et al., arXiv:2502.01224 (2025); loader + ChronoNet baseline: https://github.com/biomedepi/seizeit2	Annotations in `*_events.tsv` (`eventType` starting with `sz`; also lateralisation, localisation, vigilance). Very long recordings: work with a subset of patients and crop around seizures for training, but evaluate on continuous data.
Images	e.g. MedMNIST (https://medmnist.com), CIFAR-10/100, STL-10, Oxford-IIIT Pets, Flowers-102 (all via `torchvision` or `pip`)	Other image datasets are possible after approval; prefer datasets that fit into Colab (a few GB at most).

The course notebooks contain small loaders for both EEG datasets (`eeg_data.py`) that you may reuse and extend.

4 Ideas for your own idea (C)

Around seizure detection: fewer channels / wearable setting (18 scalp channels vs. 2 behind-the-ear channels) • multimodal fusion (EEG + ECG + movement in SeizeIT2) • cross-centre or cross-patient generalisation, domain adaptation • personalisation with a few minutes of patient data • class imbalance (loss functions, sampling) and event-based evaluation • robustness to artifacts • efficient models for on-device detection • self-supervised pretraining on unlabelled EEG • seizure-type or vigilance-dependent analysis (SeizeIT2 annotations) • explainability (which channels/frequencies matter).
Images: augmentation strategies, transfer learning from different pretrained backbones, small-data regimes, robustness to corruptions, calibration, efficient fine-tuning.
Not enough for C: training longer, untargeted hyper-parameter changes, a single lucky seed. **Negative results are welcome** if the experiment is well motivated and well evaluated.

5 Evaluation requirements

- Fixed train / validation / test split; EEG: **split by patient or recording**, never by random windows. The test set is used once, at the end.
- Metrics that fit the task. Seizure detection: sensitivity, specificity or false alarms per hour, AUROC/AUPRC, ideally event-based sensitivity. Images: accuracy / balanced accuracy / AUROC as in the paper.
- At least **5 seeds or folds** for A, B and C; report mean $\pm$ std or confidence intervals.

- **Statistical tests** for B vs. A and C vs. B (e.g. Wilcoxon signed-rank over seeds/folds/patients, McNemar on paired predictions, or a bootstrap CI of the difference).
- Report your compute (GPU type, training time) and deviations from the original paper.

6 Deliverables

Code. A Git repository (GitHub/GitLab, link in the paper) with a README (setup, data download, how to reproduce every table and figure), an environment file, fixed seeds, configuration files and no raw data. Notebooks are fine for exploration; final results should come from scripts.

Short paper. 6 pages main text in the official NeurIPS 2025 LaTeX template (https://media.neurips.cc/Conferences/NeurIPS2025/Styles.zip), plus unlimited references. Suggested structure: abstract, introduction (problem, contributions), related work, method (A, B, C), experimental setup (data, preprocessing, splits, metrics, implementation), results (tables with mean $\pm$ std, statistical test), discussion & limitations, conclusion. Add a contribution statement for each group member and add an **AI use statement** (Section 9).

Poster. One poster per group, presented on **day 8, 11 November 2026** (3-minute pitch + questions). Format: **portrait** (size to be announced). One key message, few words, large figures, results with error bars.

7 Milestones

Day 1	Kick-off. Start forming groups.
Day 2	Group fixed; 3 candidate papers (title, link, dataset, code link).
Day 3	Paper chosen; **1-page proposal**: paper, dataset, preprocessing, split, metrics, statistical test, compute estimate, A, B (idea or full method) and the planned idea for C.
Day 5	**Check-in**: data pipeline + A running; plan for B.
Day 6	A results; B running.
Day 7	B results + C running; poster draft for feedback.
10 Nov 2026	**Submission** of short paper (PDF) + link to the code repository.
Day 8	**Poster session**, 11 November 2026.

8 Assessment criteria

Paper understanding & dataset analysis	correct summary, sensible preprocessing, documented data statistics
Baseline (A) and paper method (B)	sensible baseline, faithful implementation of the paper's idea or method, explained deviations (optional: comparison with published numbers)
Your idea (C)	motivation, implementation, ablation
Evaluation	proper splits, suitable metrics, seeds, statistical test, honest discussion
Short paper	structure, clarity, figures/tables, limitations, references
Code	reproducible, clean, documented
Poster & presentation	clear message, answers to questions

Weighting: [to be announced by the lecturer]. Individual grades may deviate from the group grade if the contribution statements or the poster session show very unequal contributions.

9 Rules

- Reusing the authors' code is allowed and encouraged — mark clearly what you wrote yourself.
- **Use of AI assistants** (ChatGPT, Copilot, Claude, …):
  - *Allowed:* understanding papers and concepts, debugging, help with code, language editing.
  - *Not allowed:* AI-generated results, figures or experiments you did not run yourself; letting an AI write whole sections of the paper.
  - *Required:* an **AI use statement** in the paper: which tool, for what, and how you checked the output.
  - Every member must be able to explain every line of code and every claim in the paper.
  - Undisclosed AI use counts as an attempt to deceive. Submissions may be checked for indications of undisclosed AI use.
- Cite every source (papers, code, datasets with their licences and required acknowledgements).